GLOBAL RENEWABLE ENERGY IN 2025

How Far Have We Come?

A data-driven overview of renewable energy’s growing share in the global energy and electricity mix, spanning hydropower, wind, and solar technologies.

Source: Our World in Data – Renewable Energy | Ember (2026); Energy Institute – Statistical Review of World Energy (2025)

Audience: Policy analysts and energy sector strategists

75% of Global GHG Emissions Still Come from Fossil Fuels

75%

of global greenhouse gas emissions from fossil fuel combustion

5M+

premature deaths per year from fossil fuel air pollution

3/4

of the global energy system dominated by fossil fuels since the Industrial Revolution

To reduce CO_2 emissions and air pollution, the world must rapidly shift toward low-carbon energy sources — nuclear and renewable technologies.

Renewables Now Supply ~14% of Global Primary Energy

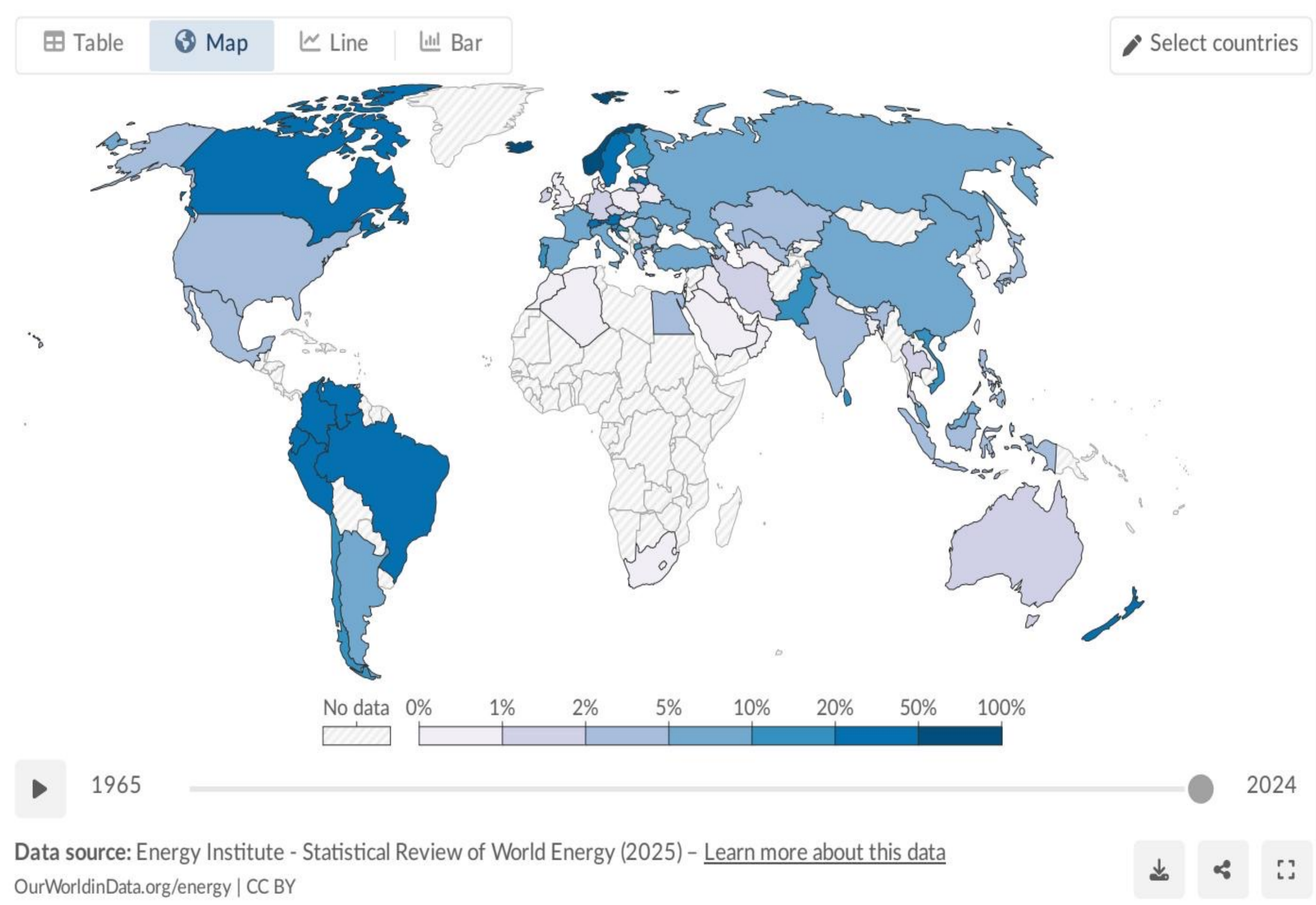
~1/7 of Primary Energy

Approximately 14% of the world's primary energy now comes from renewable technologies (2024, substitution method).

Why lower than electricity?

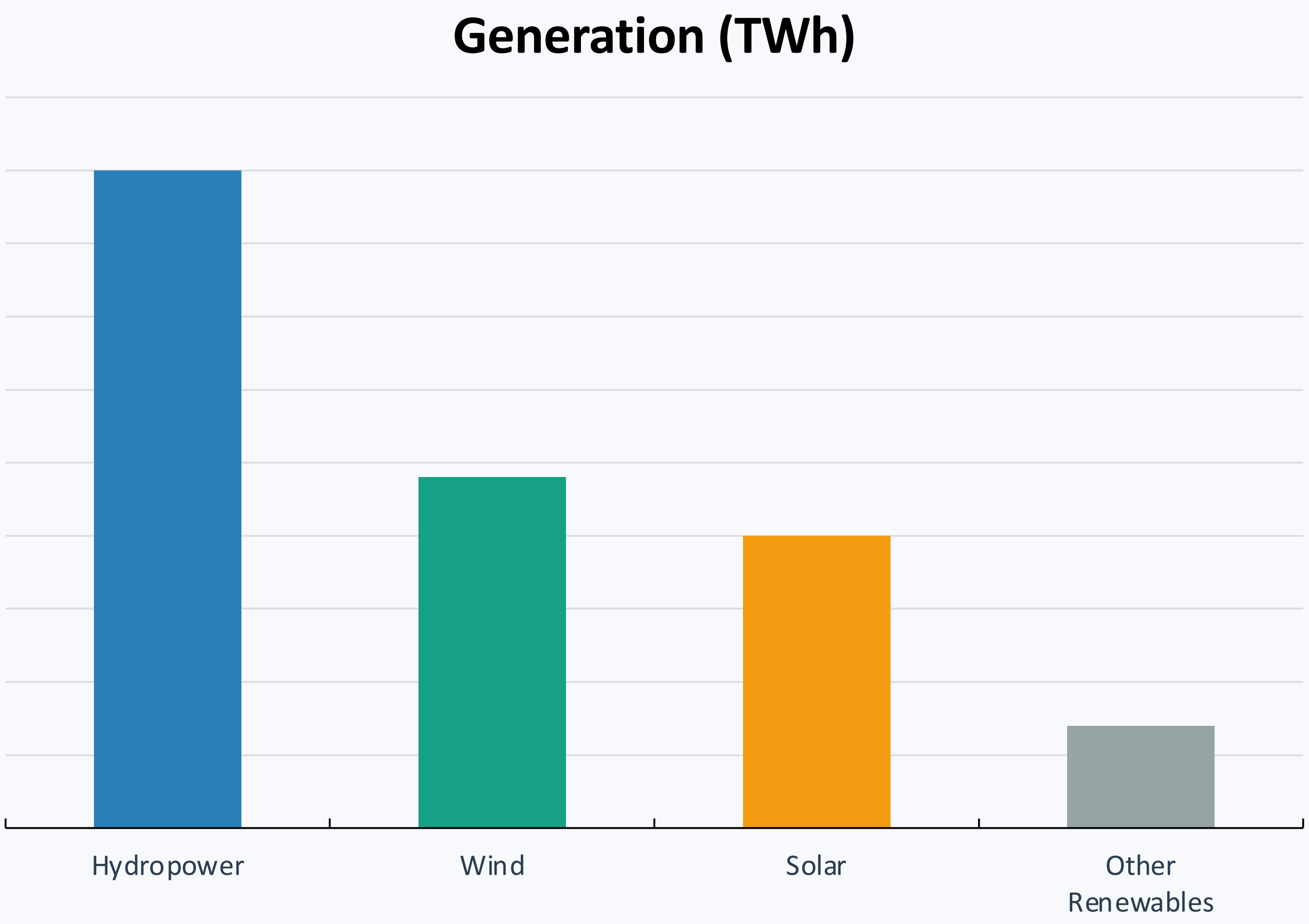
Primary energy includes transport and heating, which are harder to decarbonize and more reliant on oil and gas. The renewable share in total energy is therefore lower than in electricity alone.

Countries like Norway, Brazil, and New Zealand derive large shares from renewables; fossil-fuel-producing nations in the Middle East remain below 5%.



Source: Our World in Data – Share of primary energy from renewables, 2024

Renewable Electricity Generation Is Growing Rapidly, Led by Wind and Solar



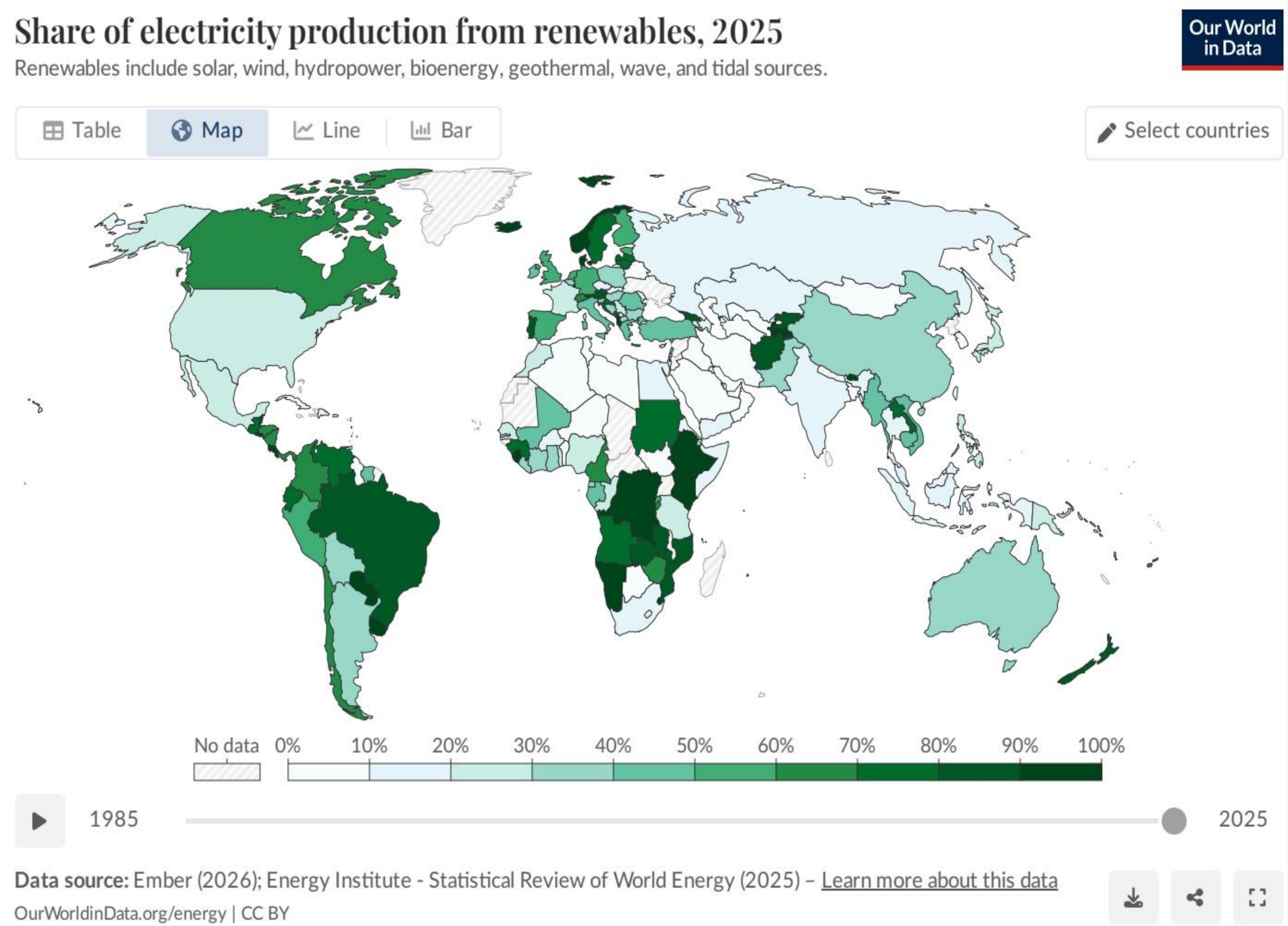
Technology Breakdown

- **Hydropower: ~4,500 TWh**
  ~47% of renewable electricity
- **Wind: ~2,400 TWh**
  ~25% of renewable electricity
- **Solar: ~2,000 TWh**
  ~21% of renewable electricity
- **Other: ~700 TWh**
  ~7% (geothermal, biomass, wave, tidal)

Wind and solar combined now nearly match hydropower's output and are growing much faster.

Data: 2024 estimates | Source: Ember; Energy Institute

Almost One-Third of Global Electricity Now Comes from Renewables



Source: Our World in Data – Share of electricity from renewables, 2025

Country / Region	Renewable Share
Norway	~98%
Brazil	~85%
Canada	~65%
Germany	~55%
China	~35%
India	~25%
World Average	~30%
Saudi Arabia	<2%

Several African and Middle Eastern countries have renewable electricity shares below 10%, reflecting fossil fuel dependence and limited investment.

Hydropower Remains the Largest Renewable Source at ~50% of Generation

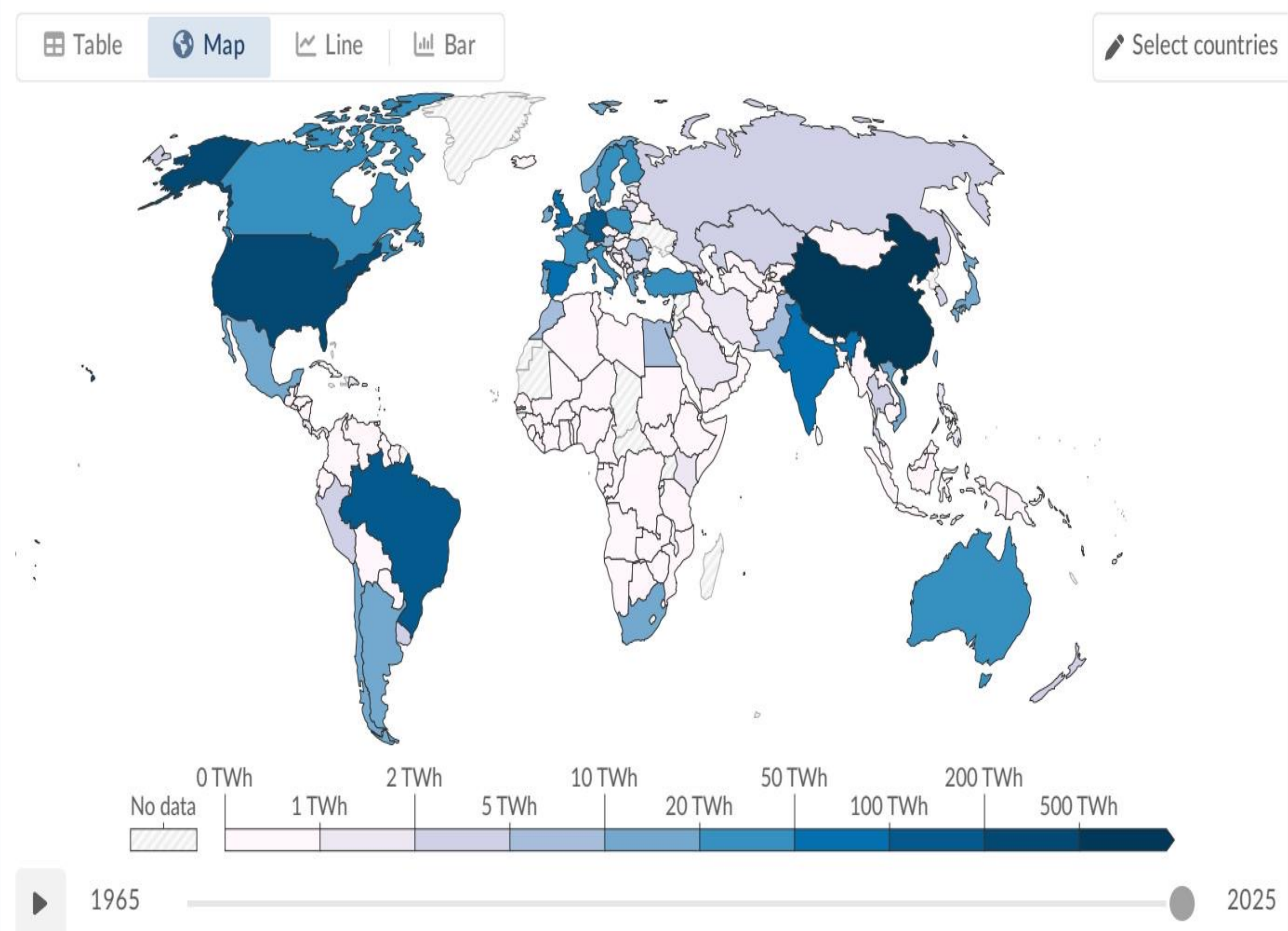
~4,500 TWh generated

~47% of all renewable electricity

Hydropower has been generating electricity at scale for over a century and remains the single largest renewable electricity source, accounting for roughly half of all renewable generation.

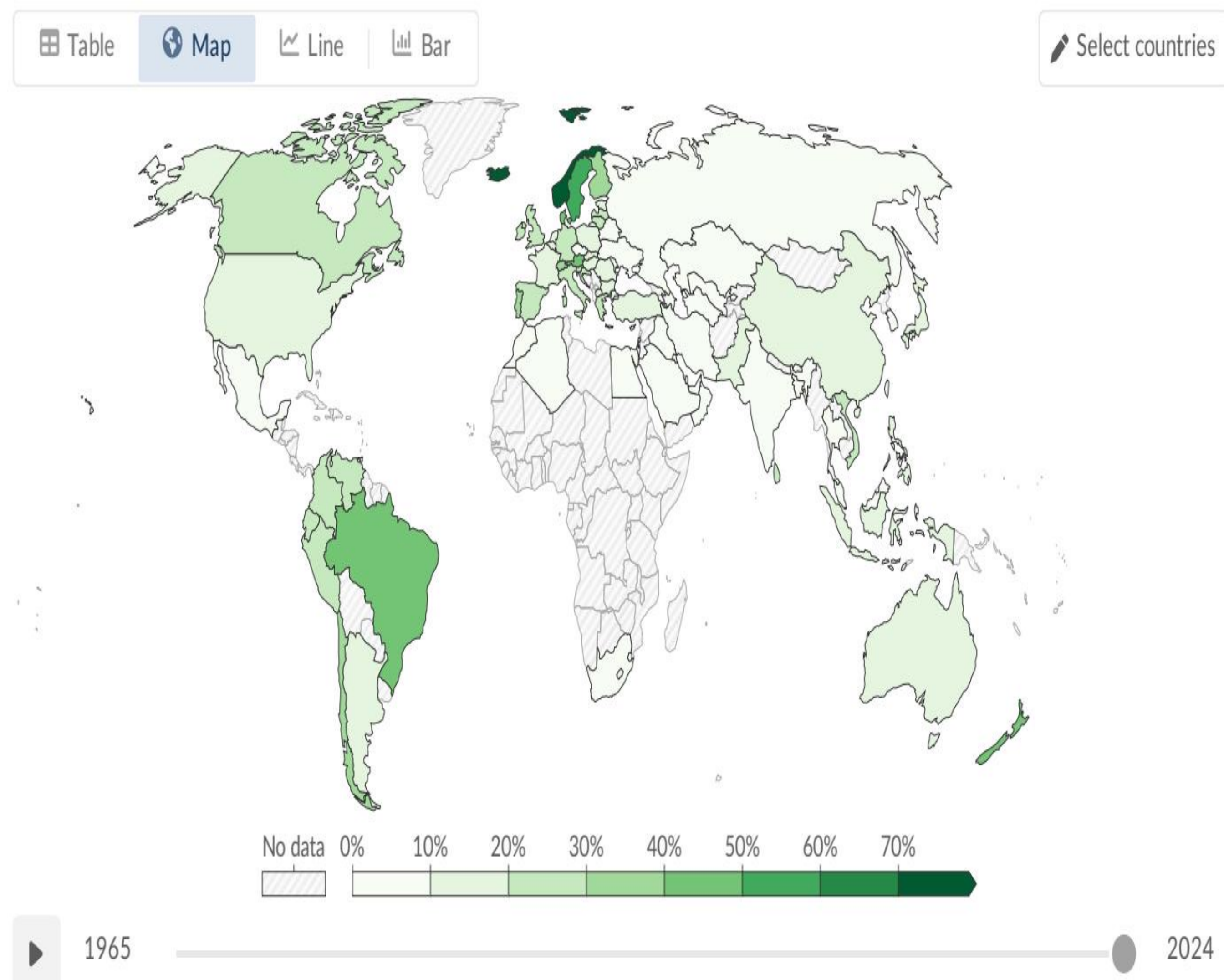
Top Producers:

- China
- Brazil
- Canada



Source: Our World in Data – Hydropower generation by country, 2025

Wind and Solar Are the Fastest-Growing Energy Technologies in History



Source: Our World in Data – Wind power generation by country, 2025

Wind vs. Solar Comparison

WIND ENERGY

~2,400 TWh | ~25% of renewable electricity

Major contributor in Europe, China, and the US. Includes both onshore and offshore installations.

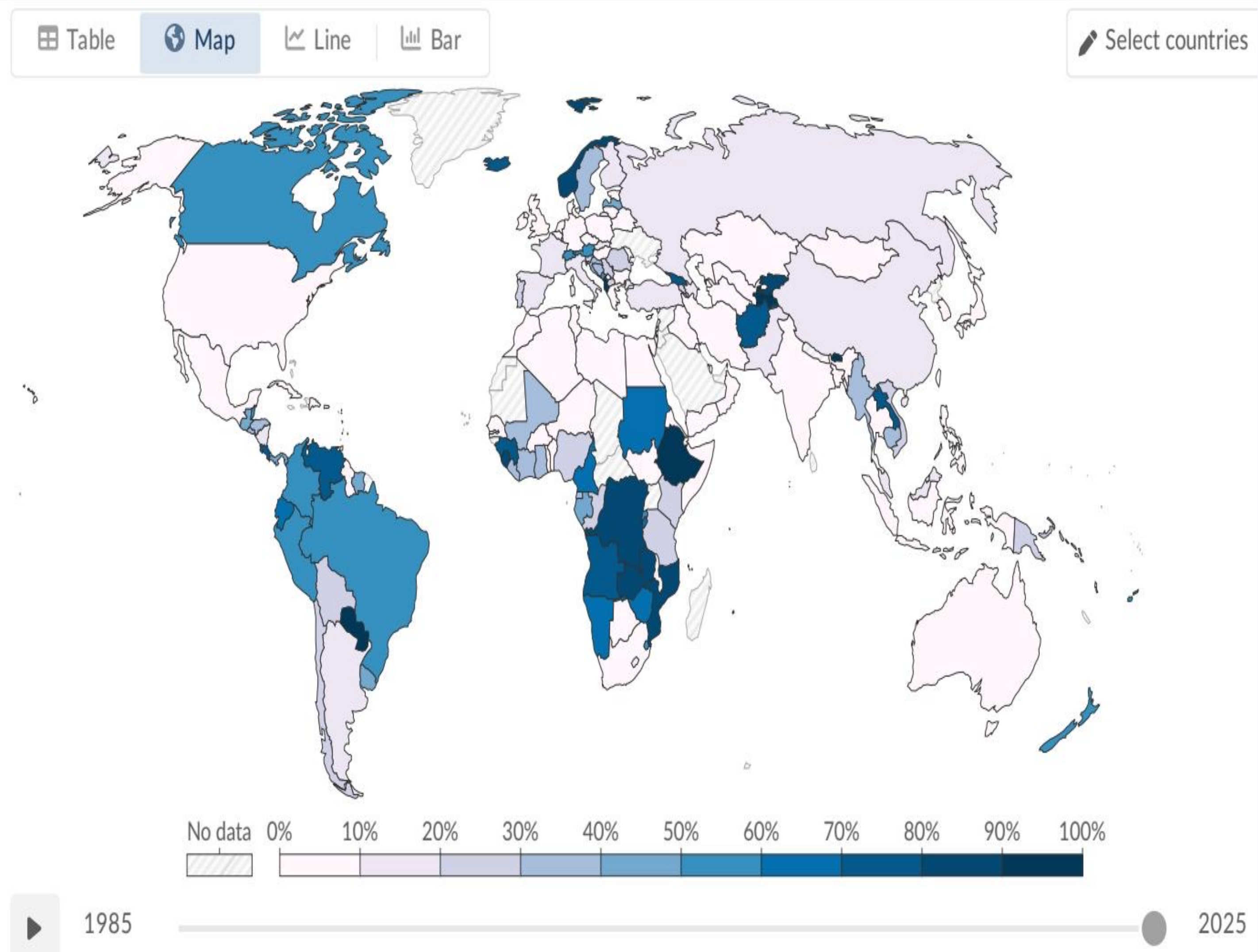
SOLAR ENERGY

~2,000 TWh | ~21% of renewable electricity

Fastest-growing source. Dramatic growth since 2010, driven by falling costs and policy support.

Combined: ~4,400 TWh — nearly matching hydropower's ~4,500 TWh output

Stark Regional Gaps: Sub-Saharan Africa Lags While Scandinavia and Latin America Lead



Source: Our World in Data – Share of electricity from hydropower by country, 2025

The Uneven Transition

⬆️ Leaders (65–98%)

Scandinavia and Brazil benefit from abundant hydropower; Denmark and Germany have invested heavily in wind and solar.

⬇️ Laggards (<10%)

Much of Sub-Saharan Africa has minimal renewable infrastructure. Middle Eastern nations remain almost entirely fossil-fuel dependent.

➡️ Emerging (25–35%)

China (~35%) and India (~25%) are scaling rapidly but still heavily reliant on coal. The US (~25%) is growing but behind Europe.

Regional disparities reflect both resource endowments and historical policy choices.

Key Takeaways: Renewables Are Growing Fast but the Transition Must Accelerate

1. **~14% of primary energy & ~30% of electricity**

   Renewables have reached a significant milestone, but remain far from sufficient for climate targets.

2. **Hydropower still dominant — but wind & solar catching up**

   Hydro generates ~47% of renewable electricity, but its growth potential is limited compared to wind and solar.

3. **Wind and solar: fastest-growing technologies in history**

   Combined, they now produce nearly as much electricity (~4,400 TWh) as hydropower (~4,500 TWh).

4. **Stark regional disparities persist**

   Scandinavia and Latin America exceed 60–90%+ renewable electricity; much of the Middle East and Sub-Saharan Africa remain below 10%.

5. **Transport and heating lag far behind electricity**

   These sectors remain heavily reliant on oil and gas, keeping the overall primary energy share much lower than the electricity share.

The renewable energy transition is underway — but to meet climate targets, the pace of deployment must accelerate dramatically across all sectors and regions.